

Basic Statistics

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Preface

1 Introduction to Statistics

- **Data:** Data consists of information coming from observations, counts, measurements, or responses. (*Ron Larson*)
- **Statistics:** Statistics is the science of collecting, organizing, analyzing, and interpreting data in order to make decisions. (*Ron Larson*)

1.1 Data Sets

There are two types of data sets you will use when studying statistics: **populations** and **samples**.

- **Population:** The entire collection of objects or outcomes about which data are collected.
 - A **parameter** refers to a numerical quantity computed from a population.
 - **Sample:** A subset of the population containing the observed objects or outcomes and the resulting data.
 - A **statistic** (rather than the field of “statistics”) refers to a numerical quantity computed from sample data (e.g., the mean, the median, the maximum).
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1.2 Branches of Statistics

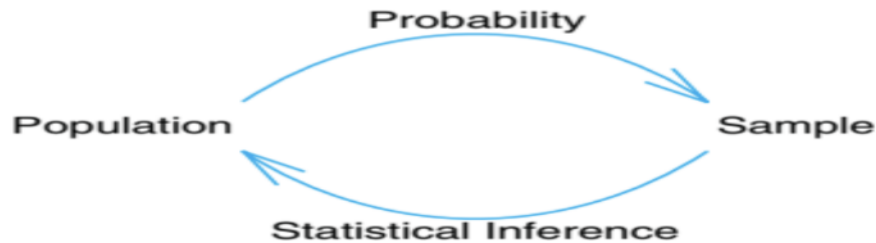
The study of statistics has two major branches:

1. **Descriptive Statistics:** Describes the important characteristics/properties of the data using measures of central tendency (mean, median, mode) and measures of dispersion (range, standard deviation, variance). Data can be summarized and represented using charts, tables, and graphs.

2. **Inferential Statistics:** Uses data from a sample to make inferences about the larger population from which the sample was drawn. The goal is to draw conclusions from a sample and generalize them to the population using probability theory.

Common methodologies include hypothesis tests and Analysis of Variance (ANOVA).

Figure 1 shows the fundamental relationship between probability and inferential statistics:



1.3 Application of Probability and Statistics in Engineering

- **Electrical Engineering:** Signals and noise are analyzed using probability theory.
- **Civil, Mechanical, & Industrial Engineering:** Used to test and account for variations in materials and goods.
- **Chemical Engineering:** Used to assess experimental data and control/improve chemical processes.

1.3.1 Why Textile Engineers Study Statistics

Textile engineers study variations occurring from material to material, test to test, sample to sample, machine to machine, time to time, and place to place. Key questions include:

1. How many tests are to be carried out to get the desired results?
2. How to analyze the collected data?
3. How to interpret the results of the analysis?

1.4 Types of Variables / Data

Data sets can consist of two types of data: **Qualitative (Categorical)** and **Quantitative**.

1.5 Qualitative (Categorical) Variables

Consist of attributes, labels, or non-numerical entries. Values are intrinsically non-numeric. *

Examples: Cause of death, nationality, race, gender, severity of pain (mild, moderate, severe).

1.6 Quantitative Variables

Consist of numerical measurements or counts where arithmetic operations make sense. *

Examples: Number of defective items, systolic blood pressure, height, age.

1.6.1 Sub-divisions

- **Discrete Variables:** Have a set of possible values that is finite or countably infinite, with gaps between possible values. Often integer values/counts. (*e.g., number of pregnancies, shoe size*)
 - **Continuous Variables:** Can take any value within an interval of the real line with no gaps. (*e.g., duration of a seizure, BMI, height*)
-

1.7 Data Collection and Experimental Design

Guidelines for Designing a Statistical Study

1. Identify the variable(s) of interest and the target population.
2. Develop a detailed plan for collecting data (ensure samples are representative).
3. Collect the data.
4. Describe the data using descriptive statistics.
5. Interpret data and make decisions using inferential statistics.
6. Identify possible errors.

1.8 Observational Study vs. Experiment

- **Observational Study:** The researcher observes and measures characteristics of interest without influencing or changing existing conditions.
- **Experiment:** A treatment is deliberately applied to part of a population (treatment group) and responses are observed. A control group (often given a placebo) is used for comparison.

1.9 Data Management & Arrays

Data management refers to processing steps occurring after collection but before analysis. Statistical software operates primarily on rectangular arrays where **rows** = **cases** and **columns** = **variables**.

Example: Medical Expenditures Data Set

Table 1.1: Typical Rectangular Data Set

ID	Age	State	Expenditures
132414	33	DHK	375
454543	25	CTG	1450
232134	41	RAJ	908
314132	55	KHL	3480
453255	46	CTG	2480
543545	35	CTG	8550
342342	28	BAR	320
563654	41	NOA	24350
465566	27	DHK	1850

2 Frequency distribution

1) **Ungrouped (discrete)** - Usually used to understand the distribution of count data

2) **Grouped (continuous)** - Usually used to understand the distribution of scale data

Example-1: Construct the frequency distribution for the following data of number of weaving defects observed in the 30 pieces of fabric. 4, 0, 1, 5, 2, 6, 0, 4, 4, 1, 5, 4, 5, 4, 3, 0, 2, 3, 3, 2, 6, 4, 1, 2, 1, 3, 4, 3, 2, 3.

Solution: Here, Minimum= 0 Maximum=6

So, the frequency table/distribution of number of weaving defects is:

Number of defects	Tally	Frequency
0		
1		
2		
3		
4		
5		
6		

Answer:

No. of defects	Freq
0	3
1	4
2	5
3	6
4	7
5	3
6	2

Example-2: Following are the results of 'hairiness index' obtained from 20 tests: 6.72, 6.52, 6.00, 6.85, 6.08, 6.64, 6.88, 6.91, 6.35, 6.52, 6.26, 7.20, 7.12, 6.36, 6.82, 6.53, 6.78, 6.75, 6.49, 6.62.

Construct the grouped frequency distribution by taking suitable/ appropriate class interval in exclusive method.

Solution: Here, Minimum=6.0; Maximum=7.2 ; Range, R= Maximum- Minimum=7.2-6.0=1.2;
n = 20

So, class interval, c =

$$c = \frac{R}{(1 + 3.32 * \log_{10} n)} = \frac{1.2}{(1 + 3.32 * \log_{10} 20)} = 0.23 \approx 0.20$$

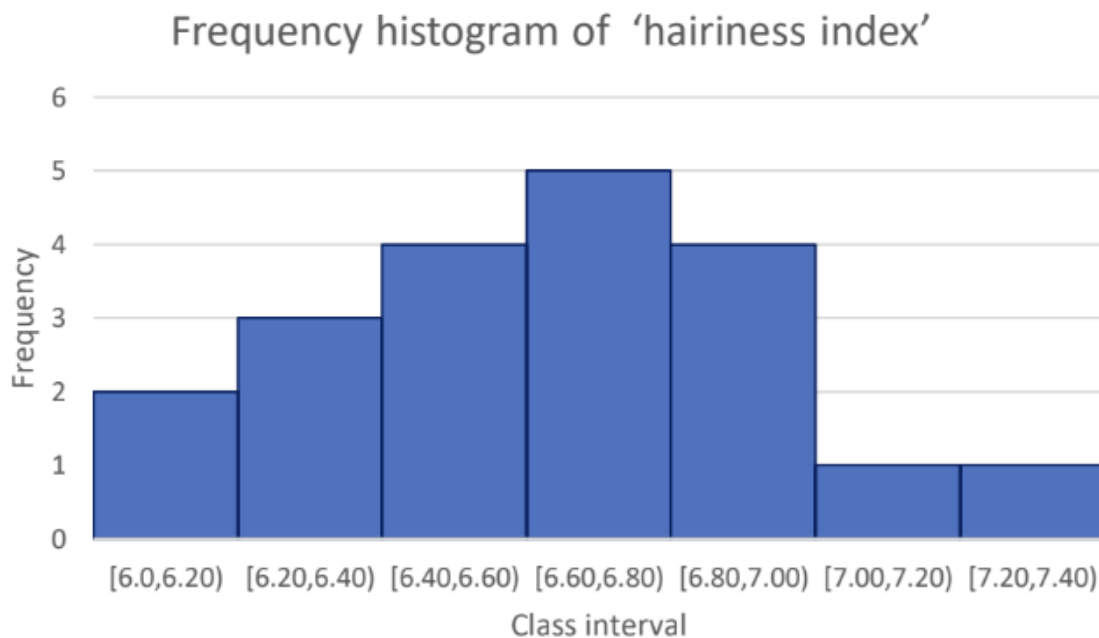
Hence in exclusive method; required classes are: [6.0,6.20), [6.20, 6.40), [6.40,6.60), [6.60,6.80), [6.80,7.00), [7.00, 7.20), [7.20, 7.40)

Frequency distribution of hairiness index (n=20):

Class interval	Tally	Frequency
[6.0,6.20)		
[6.20, 6.40)		
[6.40,6.60)		
[6.60,6.80)		
[6.80,7.00)		
[7.00, 7.20)		
[7.20, 7.40)		
Total		

Class interval	Freq
[6,6.2)	2
[6.2,6.4)	3
[6.4,6.6)	4
[6.6,6.8)	5
[6.8,7)	4
[7,7.2)	1
[7.2,7.4)	1

Histogram of 'hairiness index' data:



Histogram and shape of distribution

```
# Set up a multi-panel plot: 1 row, 3 columns
par(mfrow = c(1, 3))

# 1. Positively Skewed Distribution
# Generate data using Gamma distribution which is naturally positively skewed
# We generate a dataset of 1000 points.
set.seed(123) # for reproducibility
data_pos <- rgamma(1000, shape = 3, scale = 2)

# Create the histogram
hist(data_pos,
      breaks = 20,          # Set the number of bins
      probability = TRUE,   # Plot density, not counts
      col = "lightblue",    # Bar color
      border = "white",     # Border color
      main = "Positively Skewed", # Title
      xlab = "Values",      # X-axis label
      ylab = "Density"      # Y-axis label
)

# Add a density line for emphasis
```

```

lines(density(data_pos), col = "red", lwd = 2)

# 2. Bell-Shaped (Symmetric) Distribution
# Generate data using Normal (Gaussian) distribution
# 1000 points with mean 10 and standard deviation 2.
data_normal <- rnorm(1000, mean = 10, sd = 2)

# Create the histogram
hist(data_normal,
      breaks = 20,
      probability = TRUE,
      col = "lightgreen",
      border = "white",
      main = "Bell-Shaped (Normal)",
      xlab = "Values",
      ylab = "Density"
)

# Add a density line
lines(density(data_normal), col = "red", lwd = 2)

# 3. Negatively Skewed Distribution
# Approach: Create a negatively skewed dataset by transforming a positively skewed one.
# Generate another Gamma dataset.
data_pos_temp <- rgamma(1000, shape = 3, scale = 2)

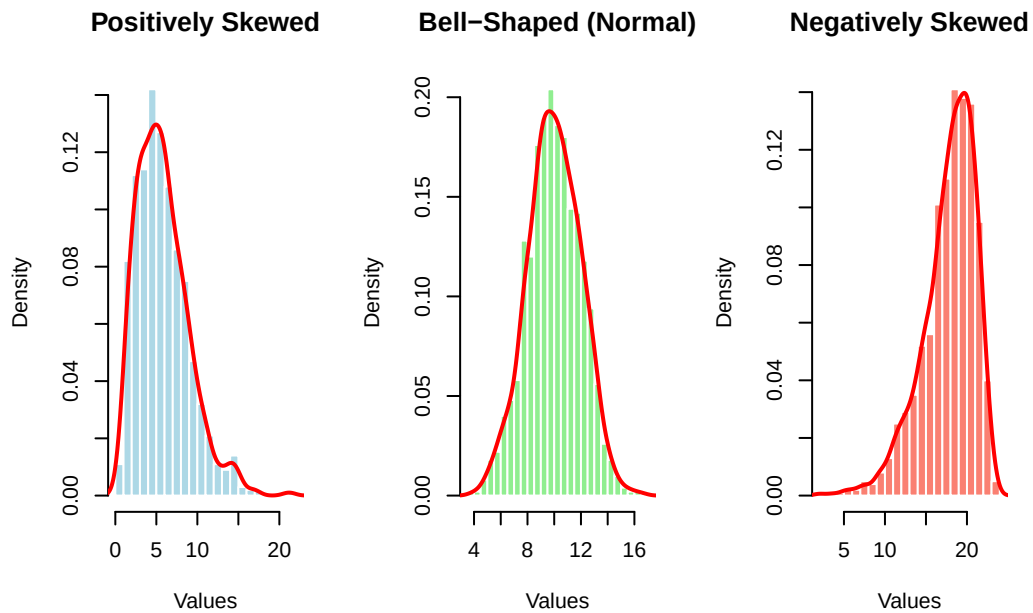
# "Flip" the data across the x-axis, then shift it.
# This results in a distribution with a tail extending to the left (negative skew).
# We choose values to keep the resulting data positive for the plot.
data_neg <- (max(data_pos_temp) + 2) - data_pos_temp

# Create the histogram
hist(data_neg,
      breaks = 20,
      probability = TRUE,
      col = "salmon",
      border = "white",
      main = "Negatively Skewed",
      xlab = "Values",
      ylab = "Density"
)

```

```
)

# Add a density line
lines(density(data_neg), col = "red", lwd = 2)
```



```
# Restore default plotting parameters (1x1 layout)
par(mfrow = c(1, 1))
```

Relative, percent and cumulative frequency distribution- example with the ‘hairiness index’ data

Class interval	Frequency	Relative frequency (rf)	Percent frequency(pf)	Cumulative frequency (cf)
[6.0,6.20)	2	0.10	10	2
[6.20,6.40)	3	0.15	15	5
[6.40,6.60)	4	0.20	20	9
[6.60,6.80)	5	0.25	25	14
[6.80,7.00)	4	0.20	20	18
[7.00,7.20)	1	0.05	5	19
[7.20,7.40)	1	0.05	5	20
Total	20	1.00	100.00	

2.1 2. Exercise

1. Construct the frequency distribution for the following data of number of weaving defects observed in the 30 pieces of fabric. 4, 0, 1, 5, 2, 6, 0, 4, 4, 1, 5, 4, 5, 4, 3, 0, 2, 3, 3, 2, 6, 4, 1, 2, 1, 3, 4, 3, 2, 3
2. Construct the ungrouped frequency distribution for the following data of number of end breaks observed on a ring frame for the 30 days. 10, 8, 5, 10, 7, 7, 11, 8, 10, 12, 11, 12, 6, 6, 7, 10, 10, 12, 10, 10, 6, 7, 8, 10, 8, 8, 12, 11, 10, 10
3. Following are the figures of number of workers absent in a textile mill every day: 2, 3, 1, 1, 2, 3, 2, 0, 0, 1, 2, 5, 0, 0, 2, 1, 1, 4, 3, 2, 0, 2, 1, 2, 2, 4, 3, 3, 3, 4 Construct the ungrouped frequency distribution for the above data.
- **4. Following are the results of 'hairiness index' obtained from 20 tests: 6.72, 6.52, 6.00, 6.85, 6.08, 6.64, 6.88, 6.91, 6.35, 6.52, 6.26, 7.20, 7.12, 6.36, 6.82, 6.53, 6.78, 6.75, 6.49, 6.62 Construct the grouped frequency distribution by taking classes 6.00–6.20, 6.20–6.40 and so on.
5. Following are the number of accidents per day observed in a textile mill during a month of 30 days: 4, 3, 3, 3, 4, 2, 2, 1, 2, 0, 2, 3, 4, 1, 1, 2, 0, 0, 5, 2, 1, 0, 0, 2, 3, 2, 1, 1, 3, 2. Construct an ungrouped frequency distribution.
- **5. Following are the number of accidents per day observed in a textile mill during a month of 30 days: 4, 3, 3, 3, 4, 2, 2, 1, 2, 0, 2, 3, 4, 1, 1, 2, 0, 0, 5, 2, 1, 0, 0, 2, 3, 2, 1, 1, 3, 2. Construct an ungrouped frequency distribution.
6. Thirty pieces of fabric were observed for the number of defects and following results were obtained. 1, 0, 0, 2, 3, 2, 1, 1, 3, 2, 4, 3, 4, 1, 1, 2, 0, 0, 5, 2, 3, 3, 4, 2, 2, 1, 2, 0, 2, 3 Construct an ungrouped frequency distribution.
7. Construct the frequency distribution for the following results of the daily sale (in 1000 Rs.) in a textile firm recorded for a month. 21, 21, 21, 22, 26, 20, 20, 20, 21, 21, 20, 20, 22, 20, 21, 23, 21, 22, 21, 25, 18, 24, 25, 19, 25, 18, 26, 23, 18, 24
- **8. Thirty linear density tests made on the yarn have shown following results. 14.8, 14.2, 13.8, 13.5, 14.0, 14.2, 14.3, 14.6, 13.9, 14.0, 14.1, 13.2, 13.0, 14.2, 13.5, 13.0, 12.8, 13.9, 14.8, 15.0, 12.8, 13.4, 13.2, 14.0, 13.8, 13.9, 14.0, 14.0, 13.9, 14.8 Construct the grouped frequency distribution by taking class intervals 12.6–13.0; 13.1–13.5 and so on.
9. Following is the distribution of fabric samples according to the strength of fabric:

Strength	150–155	155–160	160–165	165–170	170–175
No. of samples	8	12	17	10	7

Draw the histogram for the above data and interpret it.

3 Introduction to Central Tendency

Central tendency is a statistical concept used to describe the **center** or **typical value** of a dataset.

Instead of looking at every individual data point, measures of central tendency provide a single summary value that represents the entire distribution.

The three most common measures are the **mean**, **median**, and **mode**, each offering a different perspective on what constitutes the “center” of the data depending on its distribution and the presence of outliers.

3.1 Common Measures of Central Tendency

- 1) Mean / Average / Arithmetic mean
- 2) Median
- 3) Mode

Case I: Formula for Ungrouped / Raw Data

Consider a variable X with “ n ” observations that is:

$$X = \{x_1, x_2, \dots, x_n\}$$

A. Mean of X :

$$\bar{X} = \frac{x_1 + x_2 + \dots + x_n}{n} = \frac{\sum x}{n}$$

B. Median: Middle most observation of an arranged data.

C. Mode: Most frequent observation(s) in a data.

Example-1

Ten-count tests made on certain yarn have shown the following results: 22.8, 23.2, 22.9, 22.6, 23.4, 23.0, 23.1, 23.0, 22.9, 23.0

Compute mean, median and mode of the above data:

Solution:

$$n = 10; \sum x = 22.8 + 23.2 + \dots + 23.0 = 229.9$$

A) Mean:

$$\text{Mean} = \frac{\sum x}{n} = \frac{229.9}{10} = 22.99$$

B) Median: Arranged X : 22.6, 22.8, 22.9, 22.9, 23.0, 23.0, 23.0, 23.1, 23.2, 23.4

$$\text{Median} = \frac{23.0 + 23.0}{2} = 23.0$$

C) Mode:

$$\text{Mode} = 23.0 \text{ (observed 3 times uniquely)}$$

Case II: Formula for Grouped / Frequency Distribution

Consider the following frequency distribution:

Class interval	Mid value (x_i)	Frequency (f_i)
$[L_1, L_2)$	$x_1 = (L_1 + L_2)/2$	f_1
$[L_2, L_3)$	x_2	f_2
...	...	...
$[L_k, L_{k+1})$	x_k	f_k
Total		$N = \sum_{i=1}^k f_i$

A) Mean:

$$\text{Mean} = \frac{f_1 x_1 + f_2 x_2 + \dots + f_k x_k}{f_1 + f_2 + \dots + f_k} = \frac{\sum_{i=1}^k f_i x_i}{N} = \frac{\sum f x}{N}$$

B) Median:

$$\text{Median} = L_o + \frac{c}{f} \left(\frac{N}{2} - p.c.f \right)$$

Where:

L_o = Lower limit of Median class

c = class interval of Median class

f = frequency of Median class

$p.cf$ = cf (cumulative frequency of pre median class)

C) Mode: [For approximately symmetric distribution]

$$\text{Mode} = L_o + \frac{\Delta_1}{\Delta_1 + \Delta_2} \times c$$

Where:

L_o = Lower limit of Modal class

c = class interval of Modal class

Δ_1 = differences between frequencies of pre modal and modal class

Δ_2 = differences between frequencies of post modal and modal class

Example-2 Determine the mean, median and mode from the following frequency distribution and comment on it.

Length of fiber(mm)	10–15	15–20	20–25	25–30	30–35	35–40
No. of fibers	12	19	26	21	15	9

Solution:

Length of fiber(mm)	f	Mid-value (x)	fx	cf
10–15	12	12.5	150	12
15–20	19	17.5	332.5	31
20–25	26	22.5	585	57
25–30	21	27.5	577.5	78
30–35	15	32.5	487.5	93
35–40	9	37.5	337.5	102
Total	$N = 102$		$\sum fx = 2470$	

A) Mean:

$$\text{Mean} = \frac{\sum fx}{N} = \frac{2470}{102} = 24.22$$

B) Median: To determine median we have to identify the “Median” class. First calculate cumulative frequency (cf). Then, calculate:

$$\frac{N}{2} = \frac{102}{2} = 51$$

This is less than the $cf = 57$ which corresponds to class interval 20-25.

$$\text{Median} = L_o + \frac{c}{f} \left(\frac{N}{2} - p.cf \right) = 20 + \frac{5}{26}(51 - 31) = 23.85$$

C) Mode: Here the highest frequency is $f = 26$, located in the class interval 20-25. So, 20-25 is the modal class.

$$\Delta_1 = 26 - 19 = 7; \quad \Delta_2 = 26 - 21 = 5$$

$$\text{Mode} = L_o + \frac{\Delta_1}{\Delta_1 + \Delta_2} \times c = 20 + \frac{7}{7 + 5} \times 5 = 22.92$$

4 Partition values (quantiles)

5 Summary

In summary, this book has no content whatsoever.

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References